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SARDAR PATEL UNIVERSITY, VALLABH VIDYANAGAR

BCA SEMESTER-IV (NC Batch)

External Examination 2022

US04FBCA01-Computer Oriented Numerical and Statistical Method

Monday, Date: - 04-04-2022 Time: - 03:00pm to 05:00pm Marks: -70

Q.1 Multiple Choice Questions **[10]**

(1) From the following _______ method is not iterative method.

(a) False position (b) Bisection (c) Lagranges (d)none of them

(2) From the following _______ method is the best method to obtain root of equation f(x) = 0.

(a) False position (b) Bisection (c) Newton's Raphson (d)none of them

(3) The number 0.01850 x 10^3 has ________ significant digits

(a) 3 (b) 4 (c) 5 (d) 6

(4) The ________________ defines as "the art of reading between the lines of a table."
(a) Extrapolation (b) Divided difference (c) Lagrange's (d) Interpolation

(5) All the formulae of interpolation are based on the fundamental assumption that the given data can be expressed as a _______.
(a) Polynomial (b) Equation (c) Algorithm (d) None

(6) __________ method is used if the estimated value lies towards the end of the difference table.
(a) Divided difference (b) Forward difference (c) Backward difference (d) None

(7) The system of linear equation AX = B can be solved by matrix inversion method only if...

(a) $A \neq 0$ (b) $|A| \neq 0$ (c) $|A| = 0$ (d) A is symmetric

(8) The system of linear equation AX = B is said to be homogeneous if...

(a) $A \neq 0$ (b) $B = 0$ (c) $A = 0$ (d) A is symmetric

(9) Forecasts ________________________________.

(a.) become more accurate with longer time horizons (b.) are rarely perfect

(c.) are more accurate for individual items than for groups of items (d.) all

(10) Irregular variations in a time series are caused by:

(a.) lockouts and strikes (b.) epidemics (c.) floods (d.) all

(P.T.O)

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Q.2 True /False and Fill in the Blanks [08]

1. Absolute error is defined as abs (True Value – Approximate Value) State True / False.
2. For real root of an equation $x^3 - 2x - 5 = 0$, the root lies between __________.
3. The second order differences is denoted by ___________.
4. Moving Average method is not a type of interpolation method. State True / False.
5. The system of linear equation AX = B is said to be non-homogeneous if…B is equal to 0. State True/ False.
6. Rate of change of distance with respect to time represents____________.
7. The moving averages in a time series are free from the influences of________.
8. In ratio to moving average method for seasonal indices, the ratio of an observed value to the moving average remove the influence of trend and cyclic both. State True /False.

Q.3Write a short answer for given questions [20]

1. Define Relative error and absolute error.
2. Describe the stopping rules to obtain approximate solution for given non-linear equations.
3. Use the False Position method to obtain approximate solution of the equation $x^3 - 9x + 1 = 0$.
4. Draw forward difference table.
5. Draw backward difference table.
6. Explain divided difference method.
7. List only various direct and iterative methods.
8. If x lies in the upper half of the table and if $x_k < x < x_{k+1}$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
9. If x lies in the upper half of the table and if $x = x_k$, then what is $\frac{dy(x)}{dx}$ and $\frac{d^2y(x)}{dx^2}$?
10. What is Time Series?
11. List the component of Time series.
12. Write merits and demerits of 'Ratio to Moving Average' method

Q.4 Long Questions Answers (4 out of 8) [32]

1. Bisection method (up to 3 decimal place) $X^3 - X^2 - 1 = 0$
2. False position method (up to 3 decimal places) $X^3 - 2X - 5 = 0$
3. Estimate the expectation of life at the age of 16 years by using the following data:

Age (In Years)	10	15	20	25	30	35
Expectation of life (In Year)	35.4	32.3	29.2	26.0	23.2	20.4

4. If L_x represents the numbers living at age x in a life table interpolate, by using newton's method of L_x for the values of x = 24 and x= 29.
$L_{20} = 512$, $L_{30} = 439$, $L_{40} = 346$, $L_{50} = 243$

5. Solve the following system of equations using matrix inversion method

$$2x_1 + 3x_2 + 4x_3 = 20$$
$$4x_1 + 2x_2 + 3x_3 = 17$$
$$x_1 + 4x_2 + 2x_3 = 17$$

Given the following table

6.

x	0	1	2	3	4	5	6
y	0	2	2.75	3.4	3.2	2.5	2.2

Use forward difference formula to compute $\frac{dy}{dx}$ at 2.25

7. Given the following quarterly sales figures in thousands of rupees for the years 1966-1969, find the specific seasonal by the method of ratio to moving averages.

Year	I	II	III	IV
1966	290	280	285	310
1967	320	305	310	330
1968	340	321	320	340
1969	370	360	362	380

8. Using simple average method, calculate the seasonal index number for each of the four quarters for following data.

Year	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
1980	106	124	104	90
1981	84	114	107	88
1982	90	112	101	85
1983	76	94	91	76
1984	80	104	95	83
1985	104	112	102	84

—x—

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